

Recommend Me First: How the Relative Ordering of Recommendations Can Lead to Unfair Outcomes for Content Creators

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Social media platforms provide millions of professional content creators with sustainable incomes. This makes recommender systems central to how visibility and revenue are distributed. As for job markets, it is therefore important to ensure these systems distribute exposure fairly. Prior work finds, however, that popularity-based recommendations are unlikely to yield even individually fair outcomes (i.e., where the resulting popularity of creators matches their quality of content) and calls for targeted solutions. To this end, our work introduces *ordered pairwise comparison* (OPC) as a cold-start fair exploration intervention within two time-steps that ensures equal exposure across creators both in number and in recommendation order. We prove analytically that OPC yields individually fair outcomes for all creators after two time-steps. While subsequent recommendations inevitably reintroduce some bias, our simulations show that the fairness gains persist long-term with fair outcomes remaining substantially more likely than without the intervention. These gains come without any loss in user satisfaction. Altogether, our results suggest that parity in exposure counts during exploration is not enough to ensure fair outcomes for creators and that accounting for the relative order in which creators are presented is a promising intervention.¹

Keywords: recommender systems, individual fairness, cold-start problem, agent-based modelling

Reference Format:

Salima Jaoua, Nicolò Pagan, Anikó Hannák, and Stefania Ionescu. 2026. Recommend Me First: How the Relative Ordering of Recommendations Can Lead to Unfair Outcomes for Content Creators. In *Proceedings of Fifth European Conference on Algorithmic Fairness (ECAF'26)*. Proceedings of Machine Learning Research, 7 pages.

1 INTRODUCTION

For many professionals, visibility and revenue online are heavily influenced by recommender system (RS) algorithms. As a consequence, the fairness of recommendation outcomes has emerged as a major concern, with implications for multiple stakeholders. We focus on the notion of *individual fairness* (IF) for creators: items with similar attributes receive similar audience engagement [10, 32]. In the case of objective attributes such as quality,

¹An extended version with full proofs and experiments on the MovieLens dataset is available at <https://arxiv.org/abs/2510.20698> [16].

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ECAF'26, September 02–September 04, 2026, Ghent, BE

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doing so helps remunerate professionals according to the quality of their work. Ultimately, this encourages them to continue posting good content and reinforces their trust in the system.

Achieving IF, however, is difficult even in simple settings and even when the RS itself is unbiased. Both empirical [27] and theoretical [15] work show that early random differences in popularity increase over time through social influence and recommendation mechanisms. This "rich-get-richer" dynamic persists even under fair exploration [19]. Importantly, while popularity-based recommendation can be fair in expectation across creators, the probability of a fair outcome for any given creator remains low [15]. This means that the realized outcomes experienced by the creator population remain unfair. We aim to better understand and address this phenomenon.

This extended abstract considers a simplified model introduced in prior work [15, 21] and proposes a fairness-improving intervention within this setting. The model starts from an empty platform with no prior interactions between users and creators (a new community cold-start case). This means that there are no existing biases within the data that the RS needs to adjust for, thus simplifying the analysis for the purpose of this extended abstract¹. For cold-start settings, existing methods primarily aim to efficiently learn user preferences [22, 28] and the few approaches that explicitly target fairness focus on new items exposure through post-processing interventions [12, 33]. We instead intervene at the exploration stage itself, targeting fairness in outcome rather than exposure. We also adopt a multistakeholder perspective [1], considering both creator- and user-side metrics, since a fairness intervention that improves creator outcomes at the cost of user satisfaction is unlikely to be implemented in practice. Our central insight is that the relative *order* in which creators are recommended during exploration is itself a dimension of fairness, not addressed by re-ranking or post-processing methods. This insight suggests that instead of only accounting for the position the creator was shown on (e.g., through fairness metrics based on the position-discounted effect [5, 31]), it is worth also considering which creators were shown beforehand. By targeting this relative ordering in which creators are recommended, we propose *ordered pairwise comparison* (OPC) to enhance long-term individual fairness for creators while preserving user satisfaction.

2 PRELIMINARIES

We first introduce the model and evaluation metrics. The model is grounded in network science (by modeling popularity-based recommendations via preferential attachment [2, 17]), psychology (for users' maximizing behavior [29, 30]), and empirical evidence (presenting one creator only as users primarily consider the first recommendation per session [18]). It was also validated against data from Twitter and Twitch [21]. Its simplicity is deliberate: following agent-based modelling methodology, we start from a simple, tractable model [11], which lets us isolate why unfairness arises and where to intervene effectively. We later extend it to more complex scenarios.

The model most closely captures a community of experts agreeing under full information on the ordinal evaluation of creators (i.e., users having *homogeneous preferences*). We adopt this controlled starting point and later extend to noisy behavior. Let $CC_1, CC_2, \dots, CC_n$ be the content creators (CCs) in order of decreasing quality (CC_1 the highest quality). Notably, we define quality as an ordinal ranking, not as a nominal value. The platform consists of m users and n CCs forming a bipartite graph. We start with an empty network (a new-community cold-start problem [6, 28]) and grow the network iteratively. At each time step, (1) each user is recommended one creator, and (2) the user follows the recommended CC or not. A popularity-based RS recommends CC_i at time step $t + 1$ with a probability proportional to their current popularity: i.e., $P^{t+1}(CC_i) = \frac{1+a_i^t}{\sum_{j=1}^n (1+a_j^t)}$, where a_i^t corresponds to the

number of followers of CC_i at time t . A random RS (baseline) recommends each creator with equal probability. A user follows a recommended CC if and only if its quality is higher than any CC that the user has previously engaged with, capturing cross-session *maximizer* behavior [29, 30].

Evaluation. Following the multistakeholder perspective introduced above, our evaluation considers both creator- and user-side metrics. For creators, we focus on an outcome-based (ex-post) notion of individual fairness, as outcomes are more critical than exposure [3, 9, 14]. Unlike prior work that frames fairness as *equity of attention* [5] or exposure in proportion to relevance [31], we argue that fairness in exposure is necessary but not sufficient: these measures aggregate per-CC exposure, thus not accounting for who else the users have seen by the time this CC was shown to them. Under maximizer behaviour, two creators shown equally often at each timestep can still attract very different numbers of followers (as shown by the example in Figure 1). With this in mind, we measure fairness in outcomes. We say an outcome a^t is fair for CC_i (CC_i -fair) if CC_i is at most the i th most popular creator and *fair* overall if it is CC_i -fair for all $i = 1, \dots, n$. For users, we track the rank, in the quality ordering, of the best creator each user has followed; lower values indicate higher satisfaction.

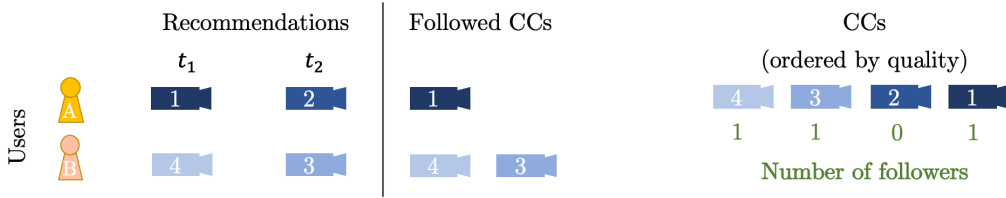


Fig. 1. Small-scale example illustrating why fairness in exposure is not enough to guarantee individually fair outcomes. The figure shows a platform with 4 content creators and 2 users: one user is shown CC_1 first, and then CC_2 , the other user is shown CC_4 and subsequently CC_3 . These recommendations could be a realization of an RS that gives equal or quality-proportional recommendation probabilities to each CC. Moreover, all creators have equal unweighted exposure, while CC_2 and CC_3 have equal position-discounted exposure. Nonetheless, this outcome under a user-maximizing behavior is unfair, as CC_3 gains more followers than CC_2 . This illustrates the need for a recommendation strategy that accounts not only for the position creators are recommended on, but also for who else is recommended beforehand.

3 AN ANALYTICALLY-INFORMED INTERVENTION

One effective yet impractical solution for guaranteeing fairness in outcomes is to consider all $n!$ orderings of creators and use each such ordering to generate recommendations for one distinct group of users. While this leads to an IF stable state in which every user has converged to have found the best CC, it requires a number of users that grows factorially in the number of creators and long search times for users (n time-steps of exploration).

This infeasible intervention reveals that ordering matters. Full symmetry across all orderings, however, turns out to be unnecessary. What is important is that each ordered pair (CC_i, CC_j) is shown to equally many users, ensuring neither creator has a systematic disadvantage from being presented after a higher-quality one. This yields a feasible alternative: consider the $n(n-1)$ ordered pairs of creators, divide users into $n(n-1)$ equally sized groups, and recommend each group the corresponding ordered pair in the first two time-steps. Later recommendations may follow any algorithm. By varying the order across groups, we ensure fairness in the chance of being followed. We call this intervention *ordered pairwise comparison* (OPC).

THEOREM 3.1. *Suppose the platform has n content creators (CCs) and $k \cdot n(n-1)$ users for some $k \in \mathbb{N}$. After the OPC in the first two time-steps, the outcome is IF for all CCs.*

PROOF 3.1. At time-step $t = 1$, we recommend each CC to exactly $(n-1)$ groups of k users, meaning $a_i^1 = k(n-1)$. At $t = 2$, users would follow the recommended creator if and only if it has better quality than the first. For each CC_i , there are $(n-i)$ pairs in the following form (j, i) where $j > i$. Hence, $a_i^2 = k(n-1) + k(n-i) = k(2n-i-1)$, which decreases as i increases. $\square$

We note that pairwise comparison proved to be effective for other applications, such as enabling better personalized recommendations [24, 26], defining a pairwise fairness metric [4], efficiently collecting ranking data in crowdsourcing [8]. Meanwhile, models such as Bradley-Terry and Plackett-Luce [7, 20, 23] infer a latent ranking from such comparison outcomes. OPC instead operates before inference, controlling which ordered pairs users see so that follower outcomes are unconfounded by presentation order. We argue that unordered comparisons do not guarantee fairness because they lack symmetry. Indeed, even when each pair is shown to equally many users, the resulting follower count may not respect the quality ranking (e.g. if CC_n is always shown first and CC_{n-1} always second across their pairs, then CC_n ends up with strictly more followers despite being lower quality).

4 SIMULATION RESULTS

To examine long-term fairness, fairness throughout the process, and user satisfaction, we simulate the model presented above². For each RS, we run the simulation with 20,000 random seeds and up to a horizon of $t = 1000$. In practice, this time limit led all simulations to converge to a state in which each user discovered the best creator (CC_1) and thus would not follow anyone new. We record the percentage of runs in which the resulting outcomes were CC_i -fair. Figure 2 shows that our intervention increases the chance of reaching fair outcomes for all creators

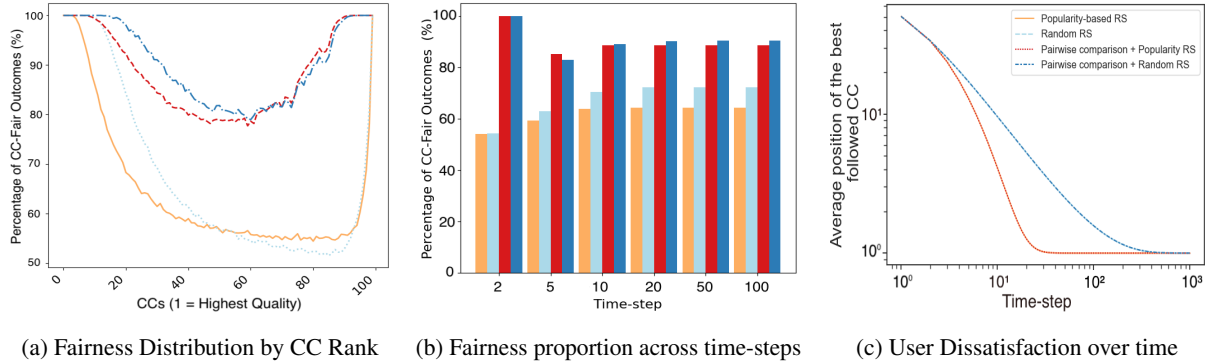


Fig. 2. Comparative analysis of recommendation systems with 100 CCs and 49500 users under four alternative RSs: random (Light Blue), popularity-based (Orange), OPC followed by random (Dark Blue), and OPC followed by popularity-based (Red). Creators are ranked by quality (CC_1 highest quality). Plots show the (a) percentage of fair outcomes per CC across simulations after 1000 time-step, (b) percentage of CC-fair outcomes per time-step averaged across creators, and (c) user dissatisfaction (position of best followed CC) across time.

²GitHub repository with the code and synthetic data: https://github.com/salimajaoua/Recommend_Me_First_ECAF26.

in the long-term. The largest differences are for mid-range quality CCs, where fair outcomes occur approximately 30% more often. The intervention also improves fairness early on, with guaranteed fair outcomes for all creators after two time-steps (aligning with Theorem 3.1). In contrast, the average chance of CC-fair outcomes for RSs without the intervention starts at only around 55%, reaching respectively 65%, 70% by time-step 100. Thus, OPC leads quickly to more fairness and maintains a higher level of IF throughout the process. Importantly, Figure 2c shows these gains do not come at the cost of users: unlike exploration-enhancing alternatives (such as uniform random recommendations, which do not rely on popularity), OPC maintains the quality of creators users see.

5 DISCUSSION

We introduce *ordered pairwise comparison*, a structured cold-start exploration to improve individual fairness that targets the order in which content creators are presented to users. Analytical results show that OPC yields individually fair outcomes after only two time-steps. We also demonstrate empirically that the resulting fairness improvements persist over time without harming user satisfaction. These results suggest OPC is a practical solution to the new-community cold-start problem and that, in general, fairness interventions for RSs are more efficient when accounting for the order of recommendations.

On the limitations side, we note these results hold under the assumptions stated in Section 2. Our goal was to understand and address content-creator unfairness in the analytically tractable setting where fairness was still shown to be a problem, leaving for future work the transferability of our learnings to more complex settings. In practice, an absolute notion of quality that ranks all creators according to the preferences of all users would not exist. While there are some dimensions (e.g., sound or video quality) and application domains (e.g., technical contributions evaluated by experts) where quality is less subjective, individual taste usually plays an important role. As such, a key focus for future work is to adapt the OPC in multi-community settings. Our results also depend on modeling users as cross-session maximizers who deterministically follow any higher-quality creator they are recommended. This behavior has only been tested for within-session behavior and only for isolated applications [18]. As such, an important next step is to design experiments to test whether cross-session user behavior indeed aligns with maximizer’s behavior.

In the extended version of this work [16], we examine OPC under broader conditions. We prove that the fairness guarantee carries over to noisy user behaviour by appropriately increasing the group sizes. We also evaluate OPC on the MovieLens dataset [13] to test how it behaves on platforms with pre-existing interaction histories. In such settings, the intervention improves fairness most strongly when deployed at early stages of a platform, with its effect attenuating as pre-existing popularity bias accumulates.

Our analysis also indicates that pairwise comparison can serve beyond the cold-start, simulated setting itself. In the future, we want to apply OPC to newly arriving users on established platforms and test whether it can help detect and correct ongoing popularity bias (similar to an A/B testing methodology [25]). Since pairwise comparison proved to be effective in real-world deployments for connected applications [4, 8, 24, 26], we expect our theoretical and simulation results to extend to real-world recommender systems.

Acknowledgments

This work was supported as a part of NCCR Automation, a National Centre of Competence in Research, funded by the Swiss National Science Foundation (grant number 51NF40_225155). The authors would also like to thank the anonymous reviewers for their helpful feedback.

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